

Solar Power Generation Forecasting Using Machine Learning

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ABSTRACT

As the world shifts towards renewable energy sources, predicting the amount of electricity generated through solar energy becomes an essential issue in running a power company. Errors in prediction of the amount of electricity from a solar source cause imbalance problems and economic damage to the energy production companies. The aim of this paper is to explore the use of machine learning models to predict AC power output in two utility-scale solar installations, based on meteorological features such as solar irradiation, ambient temperature and module temperature. Three models are used: Linear Regression as the base line model, Random Forest and Extreme Gradient Boosting (XGBoost). The models were developed using 34 days' worth of 15-minute data and evaluated using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE) and the coefficient of determination (R^2). The XGBoost model shows excellent prediction performance in the case of plant 1 but poor prediction ability in plant 2. SHAP value analysis reveal that solar irradiation is the dominant predictive feature but also shows the difference in the ranking of other less important features.

1. Introduction

The explosive growth of solar photovoltaic (PV) systems across the globe has completely transformed the operational dynamics of electric power networks. According to IEA (International Energy Agency), in 2024, solar electricity represented more than 5% of total global electricity production with an installed capacity that surpassed 1.6 terawatt globally and showed no signs of stopping. While this trend is crucial for ensuring that energy production moves away from carbon emissions, it raises another issue, which is related to the intermittent nature of solar production. This type of production cannot be regulated like thermal or hydro generation since its productivity depends on meteorological factors such as solar irradiation, temperature and cloud cover.

Accurate forecasting for utility-scale renewable energy generators is more than just an engineering luxury; it is essential for operations and economics. Consider, for instance, Statkraft, Europe's largest producer of renewable energy, with its solar parks and wind turbines scattered in various countries and climate regimes. For this kind of enterprise, operational data pipelines continuously gather data in the form of live sensor signals — power produced, irradiance measurements, and machine conditions — for all assets in their fleet. This information is aggregated by the use of internal APIs to different downstream entities like time series analysts, data scientists engaged in prediction models, and energy market traders. The accuracy and comprehensiveness of this information affect the accuracy of prediction models of different downstream companies. Forecasting errors occur from balancing costs, which include the cost associated with obtaining power or reducing power according to needs to ensure proper frequency balance of the electricity networks amounting to hundreds of millions of euros per year in Europe (ENTSO-E, 2023).

The development of machine learning (ML) algorithms shows a lot of promise when compared to physics-based or statistical methods. Machine learning algorithms do not require the creation of a physical model to understand the intricate nonlinear relationship between the inputs and power outputs. Multiple researchers have noted that algorithms like Random Forest and gradient boosting show more consistent performance than linear or autoregressive algorithms for forecasting solar energy generation (Voyant et al., 2017; Feng et al., 2019).

The present paper contributes to the literature in two ways. Firstly, the three machine learning methods used here — Linear Regression, Random Forest, and XGBoost — are compared on the task of solar power generation forecasting with the help of available sensors' data from two large utility-scale solar power plants. Secondly, a comparative analysis between the sites, showing that the same model trained with the same methods may have different performance across locations, is presented. Site-dependent nature of forecasting efficiency is crucial from the practical point of view and is often overlooked in the current literature, most of which is based only on the findings acquired with single-site experiments.

This paper is structured as follows. Section 2 will provide an overview of the existing literature on solar forecast methodologies and ML applications in the field of renewable energy sources. Section 3 will discuss the data, the pre-processing steps and the feature selection process. Section 4 will focus on the applied machine learning approach. Section 5 will report on the results. Finally, Section 6 will provide some discussion while Section 7 will conclude.

2. Research Background and Literature Review

Forecasting solar energy production has been an important field of study since the early 2000s due to rising photovoltaic penetration into the electric grid. The initial methods used were primarily physical, involving clear-sky irradiation and panel efficiency curves along with weather data from NWP models (Lorenz et al., 2009). Although physically understandable, such models suffer from the limitation of being hard to parameterize at a site level.

Statistical time-series approaches, such as autoregressive integrated moving average (ARIMA) and its seasonal counterpart, have similarly received substantial attention for use in solar forecasts. Time series analysis techniques yield relatively good performance results for short forecast periods, but they have certain limitations associated with their inherent reliance on a linear notion of time and inflexibility in using external meteorological data (Reikard, 2009; Huang et al., 2012). Such drawbacks of traditional physical models and statistical methods resulted in the emergence of machine learning methods, popularized in solar forecasting around the early 2010s.

Among the initial ML algorithms to tackle solar energy forecasting is the artificial neural network (ANN). For instance, Mellit and Pavan (2010) illustrated how well a multilayer perceptron could forecast daily solar radiation with results comparable to those produced by physics-based models. Another milestone in ANN literature on renewable energy forecasting was the early work of Kalogirou (2001), who summarized applications of ANN in this field. However, shallow ANN proved to be vulnerable to initialization, overfitting small samples, and lack of transparency.

Ensemble decision tree approaches proved especially promising. The application of Random Forest by Breiman (2001), where predictions from numerous decision trees are aggregated, was seen to significantly minimize prediction variance without increasing bias. The study by Pedro and Coimbra (2012) comparing several ML approaches for intra-hour solar

forecasting also revealed that ensemble approaches outperformed all other approaches. They pointed to the significance of choosing meteorological features, which showed that variables related to irradiance were most important — an empirical finding validated in this paper by SHAP analysis.

Gradient boosting has completely overshadowed the benchmarking of other methods in predicting tasks across different areas following the influential research of Chen and Guestrin (2016). The adoption of the approach in predicting energy is simply the natural course to follow due to its effective regularization, handling of missing values, and scaling. The method, according to Saigustia & Pijarski. (2023), outperformed the two methods, namely random forest and support vector regressions, in the prediction of solar power on a day-ahead basis in different climatic areas. Furthermore, the application of the same method in different climatic zones in day-ahead predictions shows better performance compared to neural networks in the case of predicting photovoltaic systems' energy using sensor tabular data by Jie Shi et al. (2012).

Model interpretability is becoming an increasingly essential criterion for the development of energy forecasting models. If a model provides accurate results but does not identify those variables that define its results, it is not practical for an operator who needs to understand the reason behind the obtained result or verify sensor data. The idea of using SHAP (SHapley Additive exPlanation) values proposed by Lundberg and Lee (2017) provided a consistent solution to the problem of determining the contribution of certain variables in creating a forecast based on game theory principles. Some authors have applied SHAP analysis to compare the importance of certain features identified by the model with the physical characteristics of solar power generation (Anagnostos et al., 2022; Li et al., 2021).

The impact of temperature on the efficiency of solar panels has also become an area of increasing interest. Research has found that temperature is inversely related to solar cell efficiency, resulting in a loss of around 0.4 to 0.5 percent for each degree Celsius above standard testing conditions (King et al., 2004; Notton et al., 2010). The addition of the module's temperature as a variable in ML forecasting algorithms has resulted in greater precision, especially in summer months when irradiation levels are at their peak (Cornaro et al., 2015).

One common limitation that has been evident in the current literature is that results are mostly reported from studies based on a single installation, which makes it hard to measure how

the predictive capabilities of ML-based forecasting algorithms transfer to other locations. This has been pointed out by Diagne et al. (2013), who, in their comprehensive review of various solar irradiance forecasting approaches, recommended the need for studies validating algorithms' performances across multiple installations. A study carried out by Pedro et al. (2019) comparing the performances of predictive models at different sites showed varying levels of model effectiveness, ranging from near-perfect predictions at some sites to consistently wrong forecasts due to cloud formation in others. Antonanzas et al. (2016) have shown that site-specific prediction models perform better than generic models, as long as adequate site information is present.

In addition, the economic benefits of accurate solar forecasts have been analyzed using quantitative methods. According to Bessa et al. (2015), a reduction of 10 percent in RMSE could save a significant number of percentage points in annual revenue for an operator of solar portfolio through decreased balancing cost. The study conducted by Almeida et al. (2015) presented some techniques for nonparametric photovoltaic forecasts that adjust themselves according to site-dependent generation characteristics without requiring complex physical parameters. The studies, therefore, prove that small advancements in methodology can provide tangible economic gains especially when the forecasts are made for organizations having large distributed renewable energy portfolios.

3. Materials and Data Sources

3.1 Dataset Description

In this paper, the Solar Power Generation Dataset will be analyzed using data collected and made available on Kaggle by Anikannal (2020). It includes readings from sensors at two solar photovoltaic stations in India. The readings were taken between May 15, and June 17, 2020, for 34 consecutive days. Each of the plants is represented by two different sets of data - generation data which records power output on an inverter level, and weather sensor data which records the site-level conditions during that period.

Generation data from Plant 1 and Plant 2 have around 67 thousand rows, with each row containing a record for one inverter reading at a specific timestamp. Both the plants data include DC_POWER (kilowatts), AC_POWER (kilowatts), DAILY_YIELD (total daily energy production), and TOTAL_YIELD (total lifetime energy produced) fields. Plant 1 and Plant 2

weather sensor data contains around 3 thousand rows, and the fields include AMBIENT_TEMPERATURE (°C), MODULE_TEMPERATURE (°C), and IRRADIATION (W/m²).

An important feature of the data on electricity generation is that data is collected per inverter, and a plant uses about 22 inverters. This implies that the generation files have many rows at each time stamp and require aggregation to obtain data for the plant level before joining with the meteorological data collected per sensor. It can be observed that there is no missing data in any of the four files.

3.2 Exploratory Data Analysis

Data analysis to investigate the nature and structure of the data was carried out before building models. All the main conclusions are listed below and presented in Figures 1 to 5. In Figure 1, you can see the histogram of aggregated AC power output values for both facilities. Both histograms are highly positively skewed; there is a peak at zero value due to nighttime values. The mean AC output of Plant 1 is about 6,705 kW with peak reaching up to 30,000 kW, and for Plant 2, it is about 5,012 kW with peak reaching up to 25,000 kW.

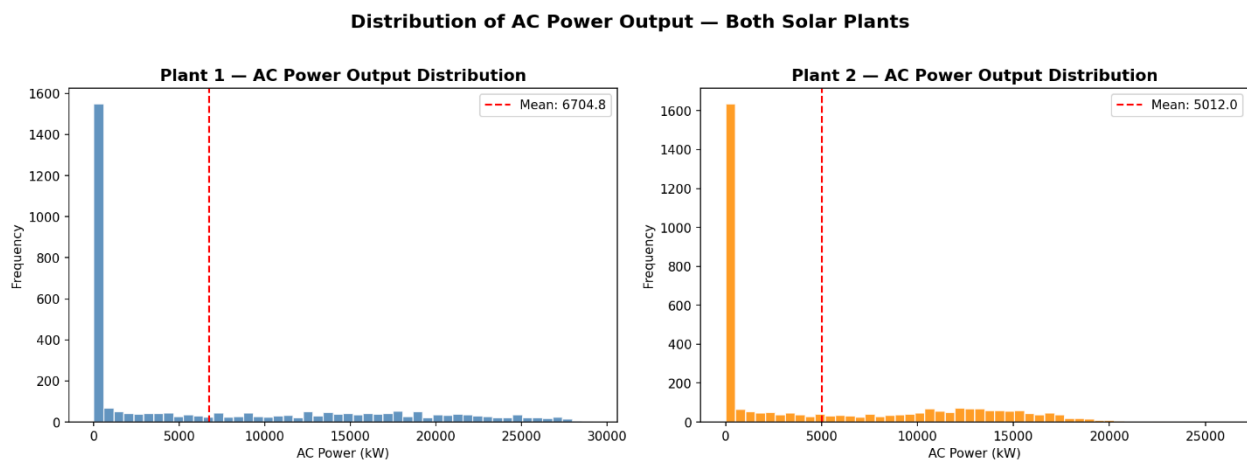


Figure 1. *Distribution of AC Power Output — Both Solar Plants (Plant 1 blue, Plant 2 orange).*

Time series for AC power production over 34 days is illustrated in Figure 2. For Plant 1, daily AC power generation trends are smooth and symmetric. Plant 2 demonstrates substantial variation during the course of the day, characterized by abrupt changes in the trend line, which

cannot be explained by variations in solar radiation. Such anomalies for Plant 2 are probably due to cloud cover events, inverter problems, and curtailment – operational problems associated with solar plants.

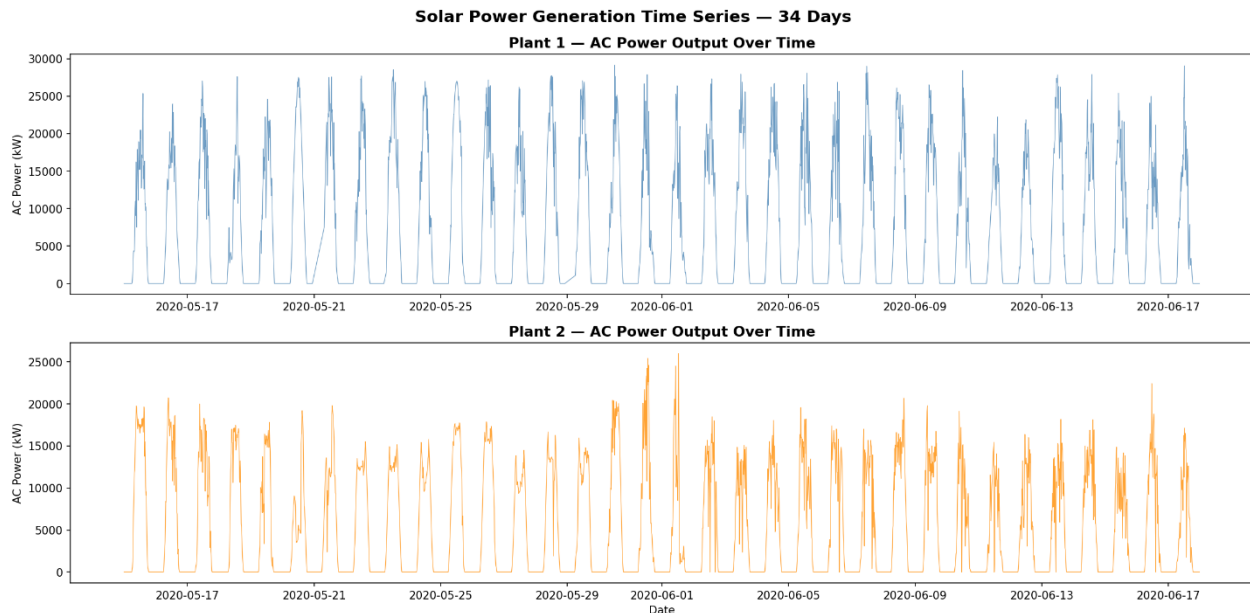


Figure 2. *Solar Power Generation Time Series — 34 Days (Plant 1 above, Plant 2 below).*

The graph on Figure 3 demonstrates the association between solar irradiation and AC power generation. It can be observed that Plant 1 exhibits an almost perfect linear correlation with a low scatter. On the other hand, Plant 2 presents a relatively higher scatter, especially when the value of irradiation exceeds 0.4 W/m^2 , where the amount of generated power varies greatly despite the same irradiance value.

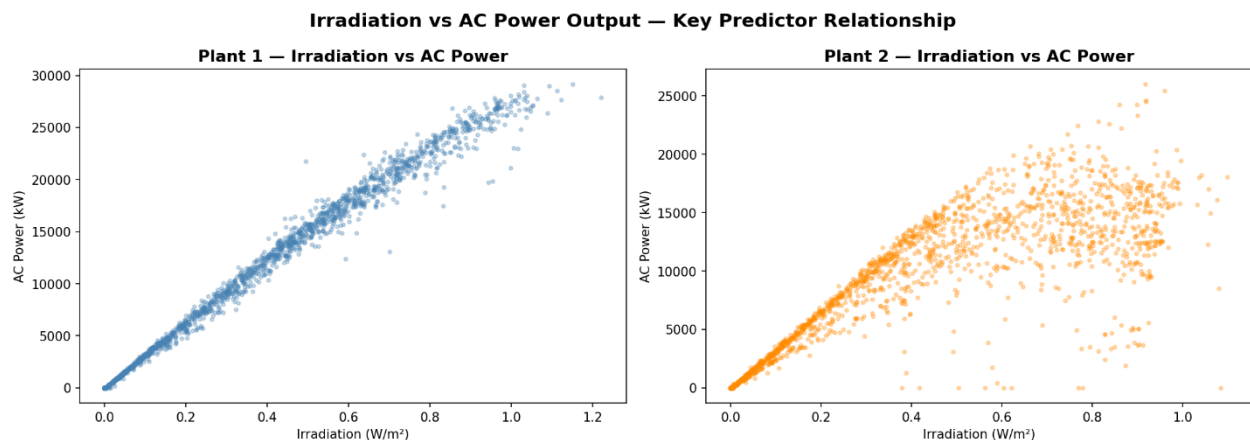


Figure 3. *Irradiation vs. AC Power Output — Key Predictor Relationship (Plant 1 left, Plant 2 right).*

Figure 4 shows that the two plant generations follow the predicted diurnal trend, having no power generated between midnight and around 6:00 AM, achieving maximum generation at noon to 1:00 PM, and back to zero generation from around 6:00 to 7:00 PM. This trend is predictable considering the sub-tropical Indian setting of these plants in the late spring season.

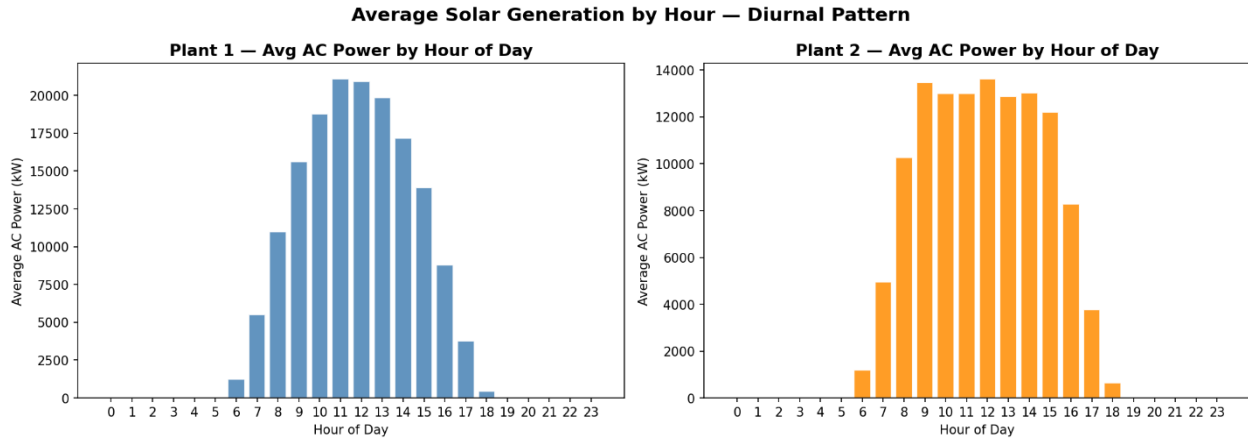


Figure 4. *Average AC Power Output by Hour of Day — Diurnal Generation Pattern*

Figure 5 contains heatmaps, that shows the correlation between the variables in the data. The correlation values of IRRADIATION and AC_POWER are 1.00 and 0.91 in Plant 1 and Plant 2, this confirms IRRADIATION is the main variable. There is a correlation between MODULE_TEMPERATURE and AC_POWER of 0.96 and 0.87. Finally, there is a perfect correlation of 1.00 between DC_POWER and AC_POWER, This means that it cannot be useful in model building.

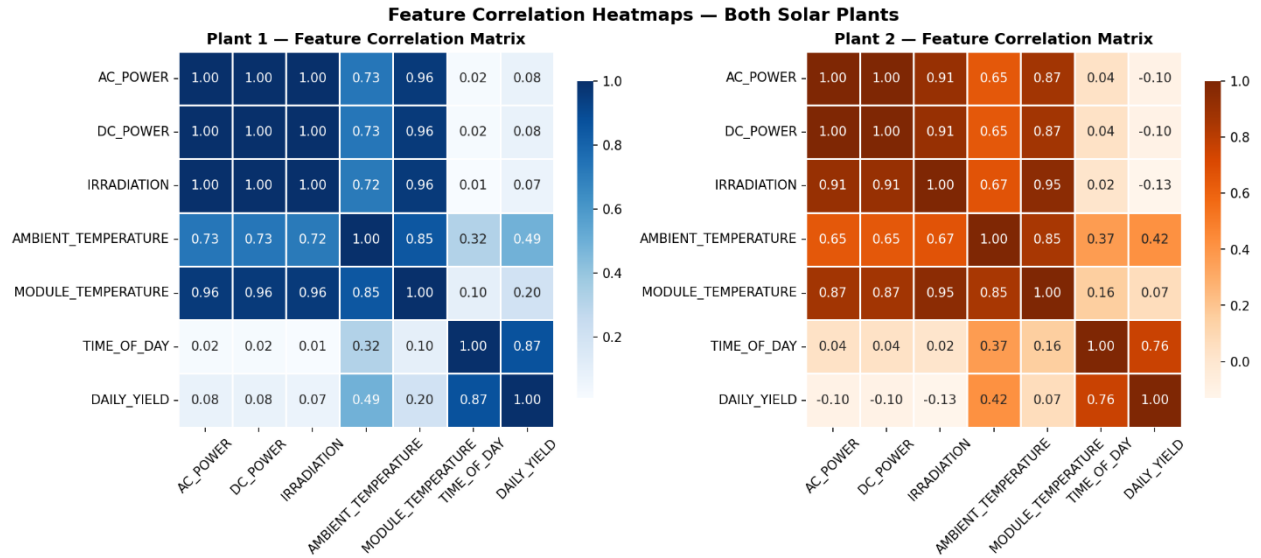


Figure 5. Feature Correlation Heatmaps — Plant 1 (blue) and Plant 2 (orange).

3.3 Data Preprocessing

The following preprocessing steps have been performed. Firstly, datetime formats have been standardized – the Plant 1 generation file had day-first format (DD-MM-YYYY HH:MM), while all others had ISO format (YYYY-MM-DD HH:MM:SS); both types have been unified into consistent pandas datetime objects. Secondly, the aggregation of generation data to plant-level has been performed by summing up DC_POWER and AC_POWER for all working inverters per one timestamp, hence reducing each generation file to ~3,200 rows per ~68,000. The third step entailed joining the aggregated generation data to weather sensor data using an inner merge based on DATE_TIME; no rows have been lost in this process. Lastly, the nighttime records (with zero irradiation) have been dropped, as they generate trivial predictions that lead to inflated metrics due to having known output in advance. This results in the following dataset sizes: Plant 1 daytime dataset size is 1,566 rows, while Plant 2 daytime dataset size is 1,641 rows.

3.4 Feature Engineering and Selection

Time-dependent features that were used from the DATE_TIME attribute include: HOUR (time in hours, expressed as integer), TIME_OF_DAY (time in continuous decimals, like 14:30 = 14.5), DAY_OF_WEEK (integer between 0 and 6), and DAY_OF_MONTH (integer between 1 and 31). There are three features that were specifically removed from the dataset to avoid potential

information leak: DC_POWER (electrical power used by the inverter to generate AC_POWER; hence not available during forecasting), DAILY_YIELD, and TOTAL_YIELD (both cumulative power generation attributes containing data about generation both in present and previous days). Final seven features used in all machine learning models are IRRADIATION, AMBIENT_TEMPERATURE, MODULE_TEMPERATURE, TIME_OF_DAY, HOUR, DAY_OF_WEEK, and DAY_OF_MONTH. The target variable in this case is AC_POWER (generated total plant AC power in kW).

4. Research Methods

4.1 Validation Strategy

Since the dataset has a temporal component, a temporal split was used for training and testing instead of a random split. The first 80% of daytime data points, ordered temporally, were used for training, and the last 20% for testing. This strategy guarantees that the test set captures a genuine out-of-sample period of the future rather than incorporating any temporal leak where future observations affect training on past observations. The testing period covers the last week of the observation period (June 11-17, 2020). Each model was fitted individually for each plant, resulting in six models altogether. Cross-pollination between plants was not considered, and one of the goals is to evaluate the performance of identical modeling techniques separately at two different locations.

4.2 Machine Learning Methods

Linear Regression is the baseline model. Linear Regression uses the technique of least squares in which the model tries to map the relationship between the feature vector in seven dimensions and AC power generated. This level of performance shows how much variance of solar output can be explained using linear feature interactions alone and provides the bare minimum level that any complex model needs to beat.

The Random Forest is a type of ensemble algorithm that creates multiple decision trees during the training process using a bootstrapped dataset with a random set of features selected for each tree. Randomness introduced by this technique ensures that each of these trees becomes uncorrelated with one another (Breiman, 2001). Random Forest algorithms had a fixed number of estimators equal to 100, and all were executed using the same random seed. Feature Importance

scores calculated using built-in functions and SHAP analysis for XGBoost models can help in measuring the importance of variables.

XGBoost creates trees iteratively, where each tree compensates for the mistakes made by the previous tree ensemble plus regularization terms based on the tree's complexity to avoid overfitting (Chen & Guestrin, 2016). XGBoost model training parameters included 200 estimators, learning rate of 0.05, and max depth of 6. XGBoost is the chosen model for interpretation using SHAP because it allows the use of TreeExplainer, which is one of the most efficient algorithms for computing SHAP values of tree-based models.

4.3 Evaluation Metrics

The assessment of the model performance is based on three metrics that complement each other. First, the root mean squared error (RMSE) is considered. This metric is penalizing high errors because of the presence of squaring in its formula; therefore, the high prediction errors can significantly impact it. Secondly, there is another metric called the mean absolute error (MAE), which is used for model performance evaluation. It can be explained as linear penalizing metric and can be interpreted as average forecast errors measured in kilowatts.

SHAP values for the XGBoost models were calculated using the Tree Explainer method. SHAP value analysis represents the contribution of each feature towards the predicted outcome by breaking down the prediction into contributions of each predictor. The SHAP contribution can be viewed as the marginal contribution of that feature when considering all subsets of features (Lundberg & Lee, 2017). The mean absolute SHAP values can be used as an important metric for measuring feature importance.

5. Results

5.1 Model Performance

The performance of all models tested on each plant is shown in Table 1. It can be observed that there is an enormous difference in the performance of models in the two solar plants.

Table 1. Model Performance Metrics — Plant 1 and Plant 2 Test Sets

Plant	Model	RMSE (kW)	MAE (kW)	R ²
Plant 1	Linear Regression	1,022.4	788.7	0.9821
Plant 1	Random Forest	656.6	424.1	0.9926
Plant 1	XGBoost	614.7	417.5	0.9935
Plant 2	Linear Regression	3,371.4	2,595.4	0.5824
Plant 2	Random Forest	2,556.0	1,401.7	0.7600
Plant 2	XGBoost	2,481.0	1,271.2	0.7738

For Plant 1, all three models achieve exceptional performance. When R^2 value is calculated for Linear Regression, we obtain a value of 0.9821, which means that there is almost a perfect correlation between the AC power generated and the variables irradiation, temperature, and time. Random Forest boosts performance, obtaining an R^2 value of 0.9926 and RMSE of 656.6 kW. Finally, XGBoost obtains an excellent result with an R^2 of 0.9935 and RMSE of 614.7 kW. The difference between Linear Regression and ensembles is pretty small.

Plant 2's performance drops significantly. Linear Regression scores $R^2 = 0.5824$, capturing less than 60% of the variance in output. Random Forest boosts it to $R^2 = 0.7600$, while XGBoost yields $R^2 = 0.7738$ – decent, although significantly lower than the performance achieved in Plant 1. While there is an increase in predictive power of linear regression when moving from Random Forest to XGBoost (0.01 for Plant 1 vs. 0.19 for Plant 2), it seems likely that the generation process of Plant 2 is more complex and exhibits significant non-linear characteristics only captured to some degree by tree-based methods. This implies the existence of other, unaccounted for variables affecting output in Plant 2.

Performance comparisons for each model on both plants can be found in Figure 6.

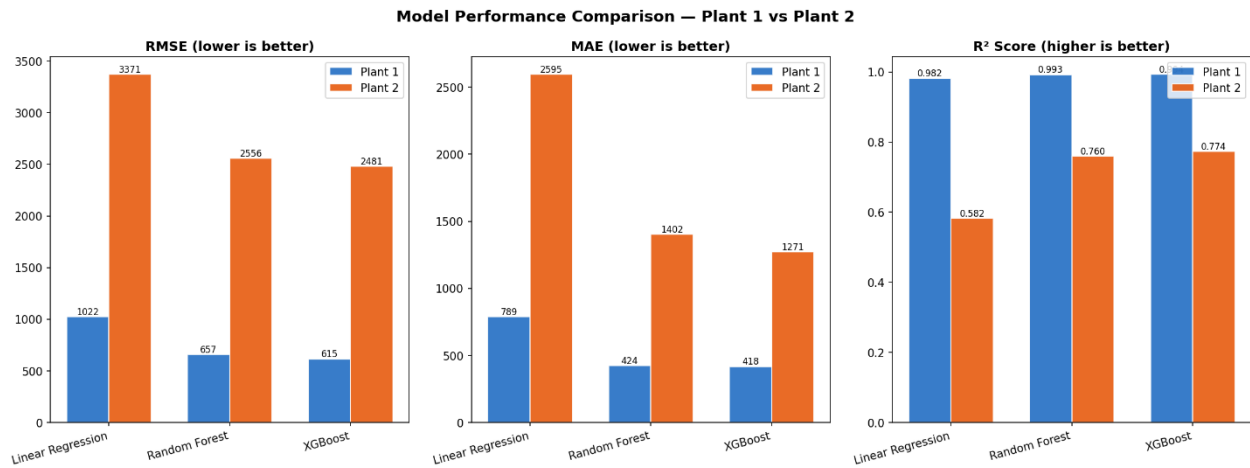


Figure 6. *Model Performance Comparison — RMSE, MAE, and R^2 for All Models and Both Plants.*

The scatter diagrams between predicted and observed values of the three models are shown for each of the plants in Figure 7 below. The predicted data of plant 1 lie close to the straight line of the perfect fit while those of plant 2 have more scatter, since the model predicts low for high powers.

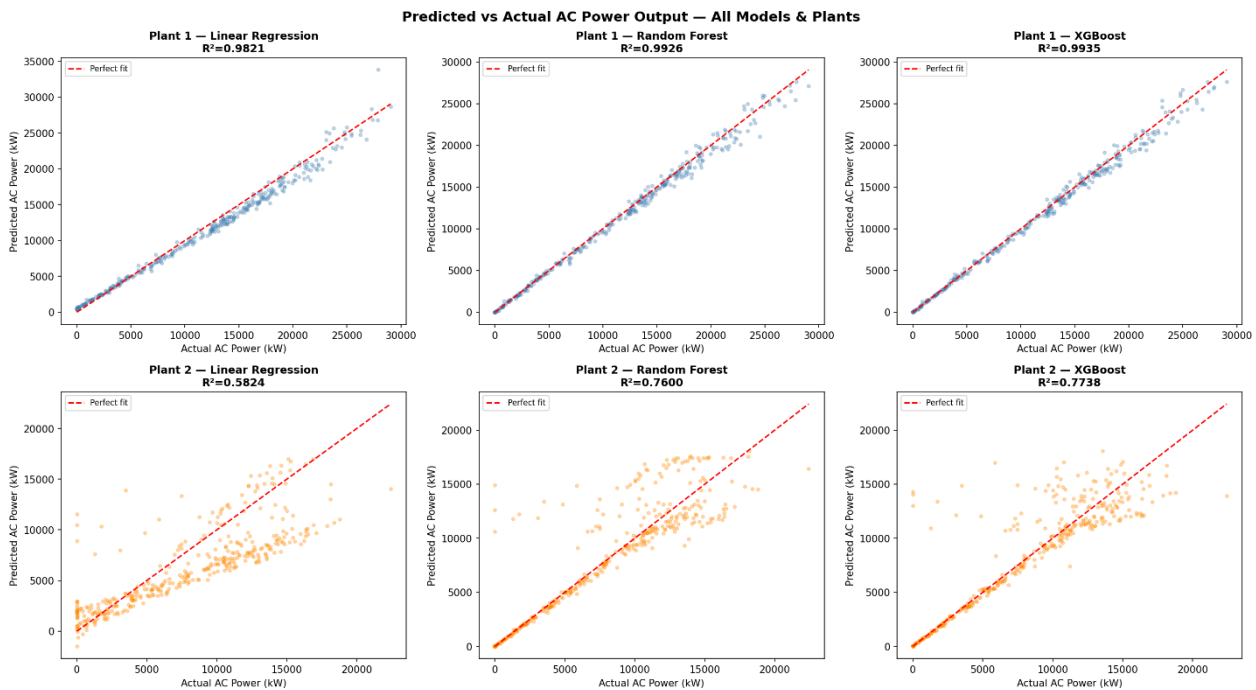


Figure 7. *Predicted vs. Actual AC Power Output — All Models and Both Plants. Red dashed line indicates perfect prediction.*

Figure 8 shows time series comparisons between the XGBoost estimated and actual values of AC power output for each plant during the testing period (June 11-17, 2020). The results demonstrate that for Plant 1, the model is highly accurate in predicting the AC power output, following the actual data closely on all seven days of the test period, even including the exact magnitude and form of each day's cycle. However, for Plant 2, while the daily cycle pattern is accurately reproduced, the model is unable to capture the rapid changes within the day. This is especially noticeable when the solar radiation is high, indicating that events not recorded by the site weather sensor may be causing the high error rates at Plant 2.

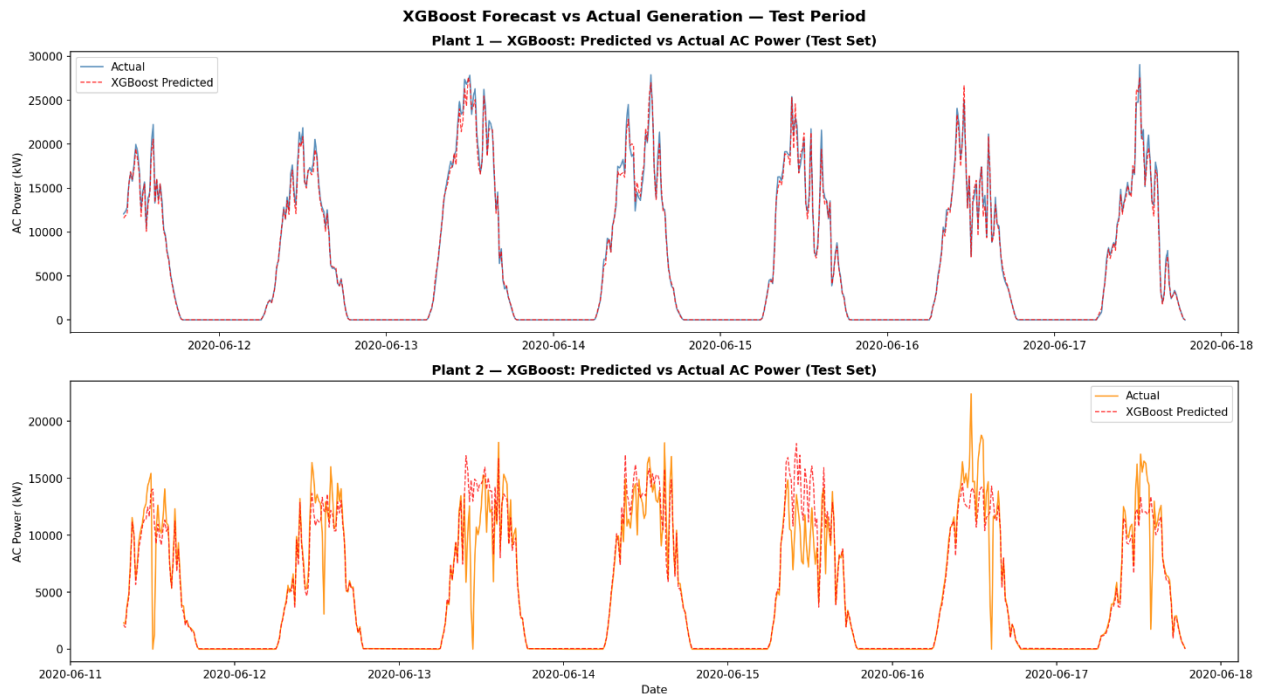


Figure 8. *XGBoost Forecast vs. Actual Generation — Test Period June 11-17, 2020 (Plant 1 above, Plant 2 below).*

5.2 Feature Importance Analysis: SHAP Values

Mean absolute SHAP feature importance scores for the XGBoost models on both plants are illustrated in Figure 9. These results confirm the physics behind solar energy generation and show some plant-specific features as well.

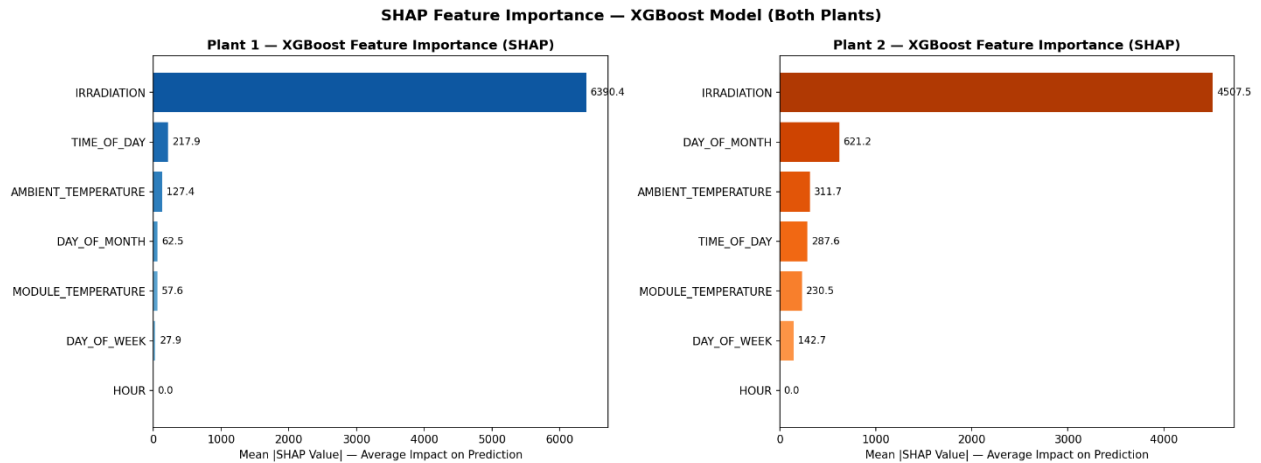


Figure 9. *SHAP Feature Importance for XGBoost Model — Plant 1 (blue) and Plant 2 (orange). Values represent mean absolute SHAP contribution in kW.*

In case of Plant 1, IRRADIATION clearly stands out as the most prominent variable with an average SHAP value of 6,390.4 kW which is more than 29 times the contribution of the next variable, TIME_OF_DAY (217.9 kW). AMBIENT_TEMPERATURE comes third (127.4 kW), while DAY_OF_MONTH and MODULE_TEMPERATURE come in as fourth (62.5 kW) and fifth (57.6 kW) variables respectively. Zero contribution of HOUR was anticipated since this aspect of time would have been captured within the continuous variable TIME_OF_DAY.

Regarding Plant 2, we can see that IRRADIATION once again emerges as the primary variable having the highest average SHAP value of 4,507.5 kW, but there are variations on how the secondary importance of other features appears in comparison to those of Plant 1. DAY_OF_MONTH assumes the second place with respect to importance as a variable (621.2 kW) and this represents nearly ten times the value observed at Plant 1 (62.5 kW).

The common conclusion that IRRADIATION is predominant in both plants is a testament to the physical consistency of the learned models. What is operationally more important is the differentiating conclusion about the importance of DAY_OF_MONTH in Plant 2. This could be attributed to changes in the level of panel dirtiness or deterioration, changes in cloudiness patterns due to the coming of monsoon season, or simply due to scheduled plant curtailment according to the calendar day.

6. Discussion

These conclusions, reached through the analysis above, have important implications in the context of developing and implementing solar forecasting systems based on machine learning techniques.

To begin with, the very high accuracy obtained from all three models for Plant 1, particularly XGBoost ($R^2 = 0.9935$), makes it clear that solar energy generation can be highly accurately predicted provided there is enough sensor data for describing the process of generating the energy. That is to say, provided there is enough data, any linear regression model – such as Linear Regression ($R^2 = 0.9821$) – will do fine, and there is no real advantage in using ensemble models.

It is the significantly poorer performance of Plant 2 which provides more insight into this phenomenon. Though the same parameters have been used in designing the models for both the sites, the model trained for Plant 2 performs poorly since it can explain only 77% of variance as opposed to the 99% of Plant 1. This difference certainly cannot be due to the use of models but must be a result of actual uncertainty inherent in prediction due to some influence unique to that location which is not covered by the features.

The SHAP values offer an additional layer of explainability above and beyond the performance measures. First, confirming that IRRADIATION is the most influential predictor at both locations assures us about the soundness of our models and allows for operational confidence. It can be expected by anyone knowledgeable of the physics of solar power generation that irradiation is the key determinant of solar power output, and the SHAP analysis confirms this intuition in respect of our XGBoost models. Second, the SHAP value associated with MODULE_TEMPERATURE is surprisingly lower than expected, even though there exists a strong correlation between this predictor and the output variable (see Figure 5). The discrepancy between SHAP values and correlation arises from the fact that once the model has incorporated the influence of IRRADIATION, the information provided by MODULE_TEMPERATURE is limited, because both variables are correlated.

6.1 Limitations

A number of limitations associated with this research must be stated. First, the 34-day time period is confined solely to the late spring to early summer months without providing an

opportunity for evaluating seasonal model extrapolation capabilities. Next, the dataset does not contain such covariates that are thought to have an effect on solar power generation apart from radiation and air temperature, namely wind speed, atmospheric humidity, aerosol optical depth, and solar panel soiling status. The latter could be included in order to decrease prediction error in cases when standard predictors provide inadequate variance reduction, such as with Plant 2. Lastly, this paper deals exclusively with nowcasting, as opposed to multi-step ahead forecasting based on numerical weather predictions.

7. Conclusion

In this study, machine learning models were used to predict solar power generation based on data obtained from sensors installed at two large-scale photovoltaic plants. Among the three algorithms utilized in this study, Linear Regression, Random Forest, and XGBoost were trained on 34 days of historical data recorded in 15-minute intervals and validated by temporal cross-validation.

The following are the key insights drawn from the analysis in this paper. Firstly, the model implemented by XGBoost is characterized by excellent predictive performance at Plant 1, where the R^2 equals 0.9935 and RMSE amounts to 614.7 kW, implying that machine learning methods have great potential for predicting solar power generation if the right meteorological inputs sufficiently capture the plant's power-generation potential. Secondly, employing machine learning models for solar power prediction leads to significantly poorer predictive accuracy at Plant 2, where R^2 drops to 0.7738 and RMSE rises to 2,481.0 kW. Thirdly, SHAP analysis reveals that irradiance is the most critical feature across both plants; however, DAY_OF_MONTH emerges as a substantially more influential variable at Plant 2 than Plant 1.

Further research needs to focus on certain shortcomings. A longer duration for observing seasonal effects would allow the evaluation of generalizability of models under various irradiation conditions. The inclusion of other operational and environmental factors such as wind velocity, humidity, curtailment status and inverter failures could help in minimizing the forecast error at plants similar to Plant 2. The multi-step-ahead prediction using NWP forecast as input will cater to the relevant forecasting time window for electricity market applications. Using transfer learning techniques based on data-rich sites for predicting energy generation at less informative sites is highly recommended.

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Appendix: AI Usage Declaration

As per the specifications of the course, this appendix is provided that highlights the use of AI tools for this project.

Claude (Anthropic, claude.ai, Claude Sonnet 4.6) was used as a secondary aid for basic initial brain storming, understanding the structure of a project paper and understanding few errors in code execution.

Author's Original Contribution:

This project is the independent work of the author. All core research components, including the selection of the research topic, the design and execution of the methodology, searching for public datasets, coding, feature engineering and the interpretation of results were performed entirely by the author without AI generation. The research questions and technical discussions are rooted in the author's professional background and domain expertise in European renewable energy and full stack application development. Specifically, experience in collection and consolidation of operational sensor data from solar and wind farms provided the necessary domain knowledge to analyze the performance differences and practical industrial implications. The author maintains full ownership and responsibility for the entirety of the work.